University of Tabuk
Faculty of Computers & Information Technology
Department of Computer Engineering

Computer Networks Lab
CEN-431

Final Project . Design and Implementation of a Multi-Service Network Using Cisco Packet Tracer 20 points

Project Overview:
This project aims to design and implement a scalable network for a fictional organization with multiple departments, integrating both wired and wireless connectivity, using dynamic routing protocols RIP and EIGRP, and providing essential network services including DHCP and DNS. The network will consist of a LAN in the main office, a WAN link to a remote branch office, and an internet connectivity simulation. In addition, VLANs will be implemented in the main office to segment departmental traffic, enhance security, and improve network manageability.

Network Design Requirements:

Main Office:

- 2 Routers (Router1 with EIGRP, Router2 with RIP for educational contrast)
- 2 Layer 2 Switches for departmental segmentation with VLAN configuration
- 1 Wireless Access Point for wireless LAN connectivity
- 4 PCs (2 connected to each switch, representing different departments)
- 2 Laptops for wireless connectivity
- 1 Server configured as a DHCP server
- 1 Server configured as a DNS server

Remote Branch Office:

- 1 Router configured with RIP
- 1 Switch
- 2 PCs

WAN Link:

- Simulated WAN link between the main office and the remote branch office

Internet Connectivity:

- Simulate internet access using a loopback interface on the main office's edge router

Configuration Tasks:

Routing Configuration:

- Configure EIGRP on Router1 in the main office to dynamically route between departments.
- Configure RIP on Router2 in the main office and the router in the remote branch office for WAN connectivity.

Switch Configuration:

- Configure VLANs on switches to segment departmental traffic (e.g., VLAN 10 for HR, VLAN 20 for Finance, VLAN 30 for IT).
- Assign switch access ports to their respective VLANs.
- Configure trunk links between switches to carry multiple VLANs.
- Implement inter-VLAN routing using Router-on-a-Stick on Router1.

Wireless Setup:

- Configure the Wireless Access Point with SSID and WPA2 security.
- Connect laptops wirelessly and ensure they receive IP addresses via DHCP.

DHCP and DNS Configuration:

- Configure the DHCP server with multiple scopes, one for each VLAN, for automatic IP

address assignment.
- Configure the DNS server with domain-name-to-IP mappings for internal services.

WAN and Internet Connectivity:
- Set up a simulated WAN link between the main and remote offices.
- Configure NAT on the main office's edge router for internet access.

Deliverables:

Cisco Packet Tracer File: A completed Packet Tracer file with the network fully implemented and annotated.

Project Report:
- Network topology diagrams for the main office, remote office, VLAN segmentation, and overall network.
- Detailed configuration steps for each protocol and service.
- Testing and verification procedures with expected outcomes.
- VLAN testing to verify inter-VLAN communication and DHCP scope assignment.
- Challenges encountered and how they were resolved.

Timeline & Submission:

Deadline: **November 15, 2025, at 11:59 PM**.

Submission: In class, using your laptop with the Cisco Packet Tracer file and the comprehensive project report.

Members responsible for this project:

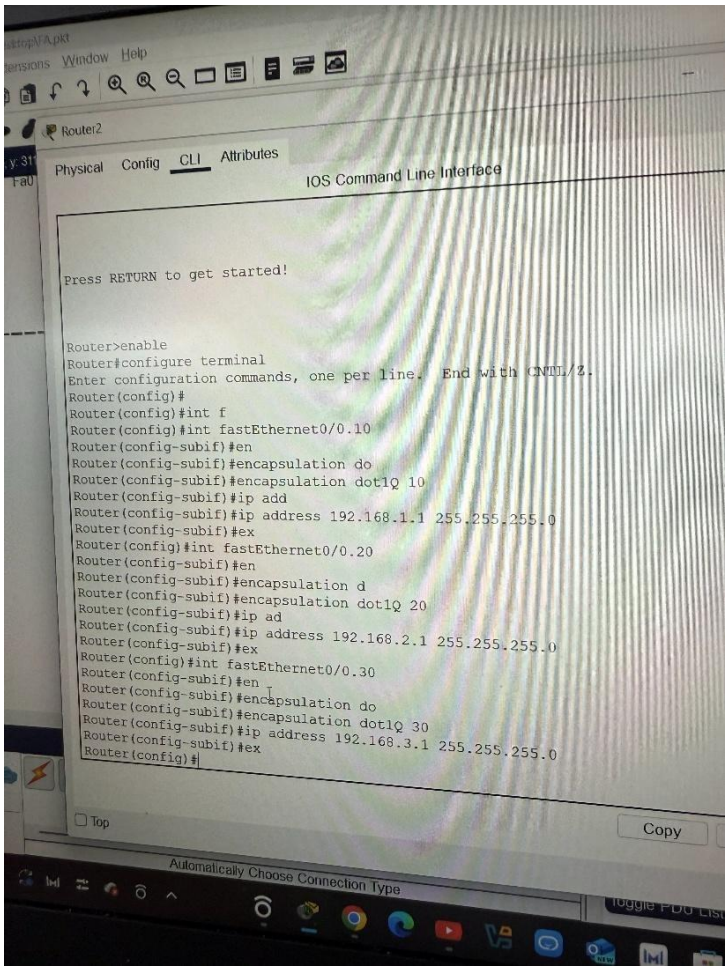
عبدالمحسن مشعان المطيري - 431001840

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فيصل فهد الشهري - 431008288

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Our work steps for this project:

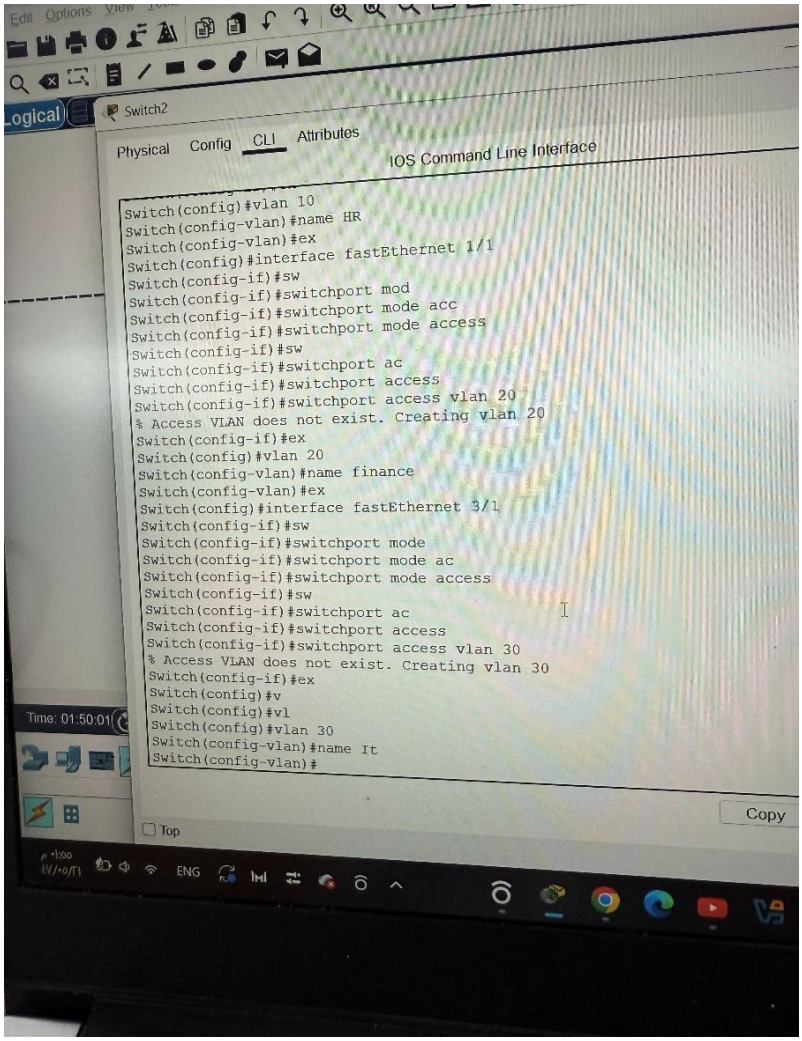


Router Configuration

The Router2 It shows the configuration of three subinterfaces on the physical interface **FastEthernet 0/0**:

- **fastEthernet 0/0.10**: Configured with **802.1Q encapsulation for VLAN 10** (encapsulation dot1Q 10) and assigned the IP address **192.168.1.1**.
- **fastEthernet 0/0.20**: Configured with **802.1Q encapsulation for VLAN 20** (encapsulation dot1Q 20) and assigned the IP address **192.168.2.1**.
- **fastEthernet 0/0.30**: Configured with **802.1Q encapsulation for VLAN 30** (encapsulation dot1Q 30) and assigned the IP address **192.168.3.1**.

This is a classic configuration for **"Router-on-a-Stick" Inter-VLAN Routing**.

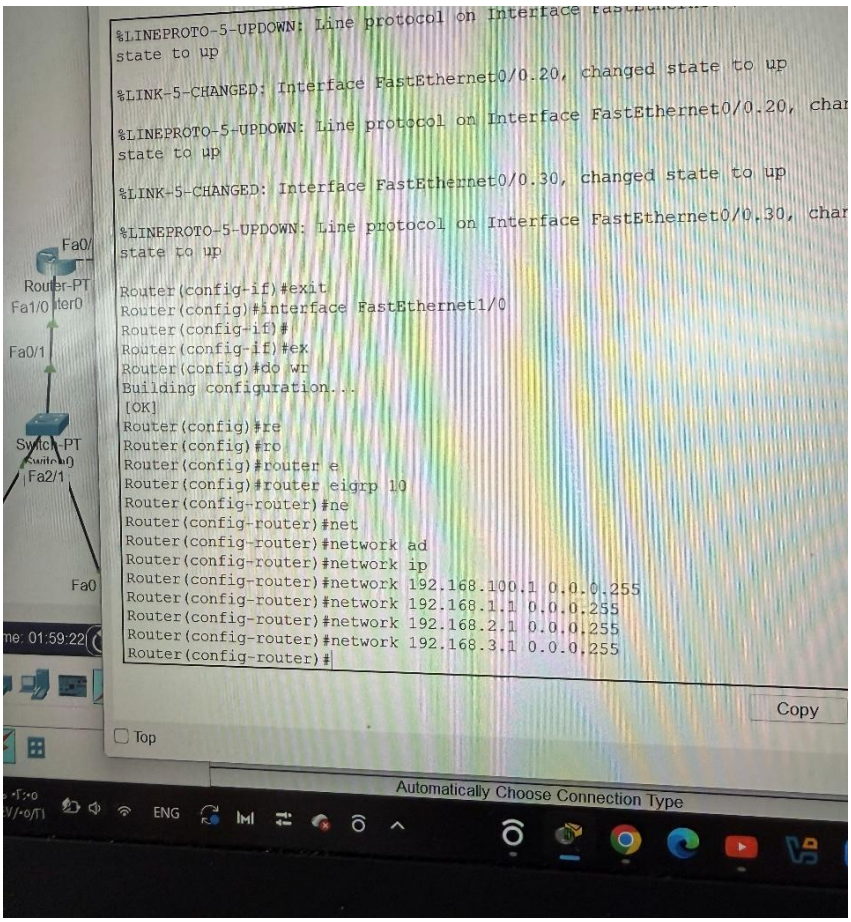


Switch VLAN and Port Configuration

The Switch2. It shows configuration steps for:

- **VLAN Creation and Naming:** Creating and naming **VLAN 10 ("HR"), VLAN 20 ("finance"), and VLAN 30 ("It")**.
- **Port Assignment:** Configuring interfaces **fastEthernet 1/1** and **fastEthernet 3/1** as **access ports** and assigning them to **VLAN 20** and **VLAN 30**, respectively.

This process isolates network traffic into different logical broadcast domains on the switch.

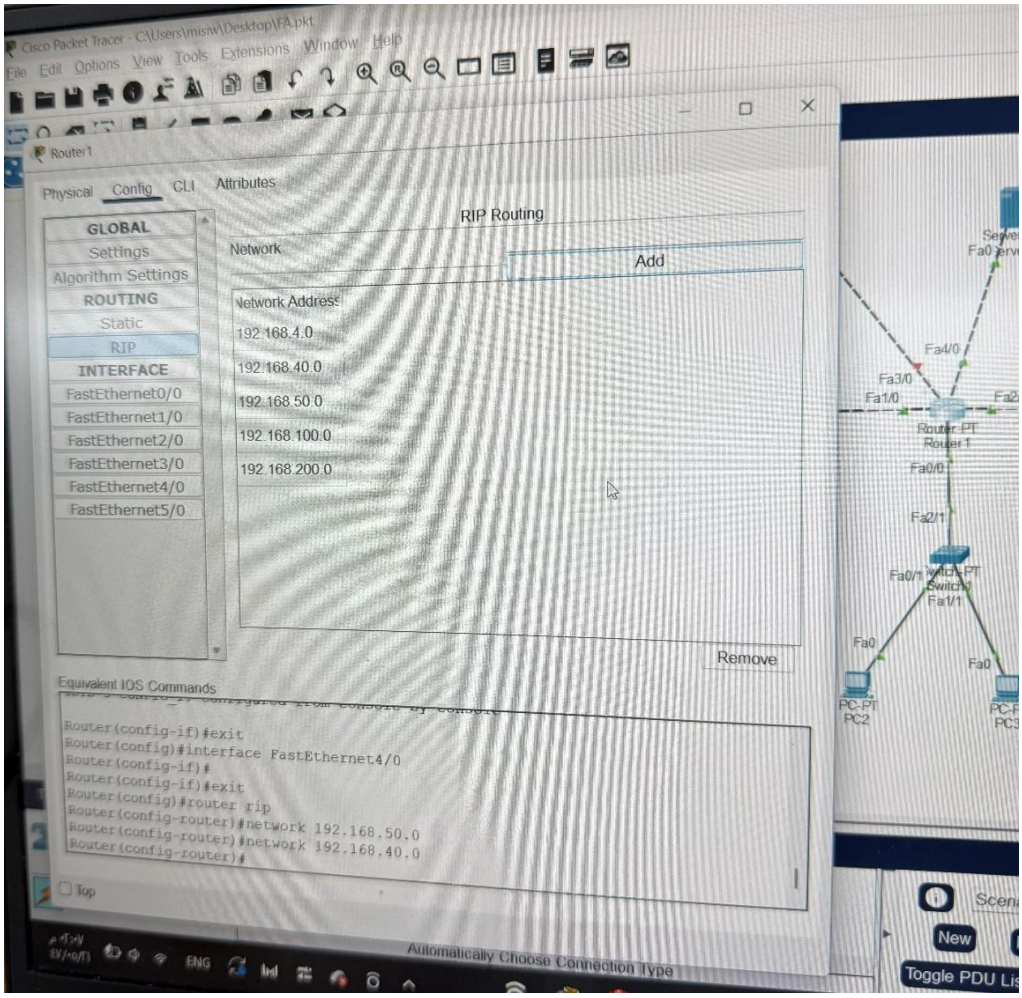


Router Interface and EIGRP Configuration

The router, showing two main activities:

1. **Interface Status Changes:** Messages at the top indicate that the line protocol on subinterfaces **FastEthernet0/0.20** and **FastEthernet0/0.30** has successfully changed state to "up".

2. **EIGRP Configuration:** The commands show the activation of the **EIGRP routing protocol (Autonomous System 10)**, followed by the configuration of several **network commands** that enable EIGRP on the corresponding interfaces:
   - 192.168.100.1 0.0.0.255 (likely a direct interface IP, often used in older EIGRP configs)
   - 192.168.1.1 0.0.0.255 (VLAN 10 gateway)
   - 192.168.2.1 0.0.0.255 (VLAN 20 gateway)
   - 192.168.3.1 0.0.0.255 (VLAN 30 gateway)



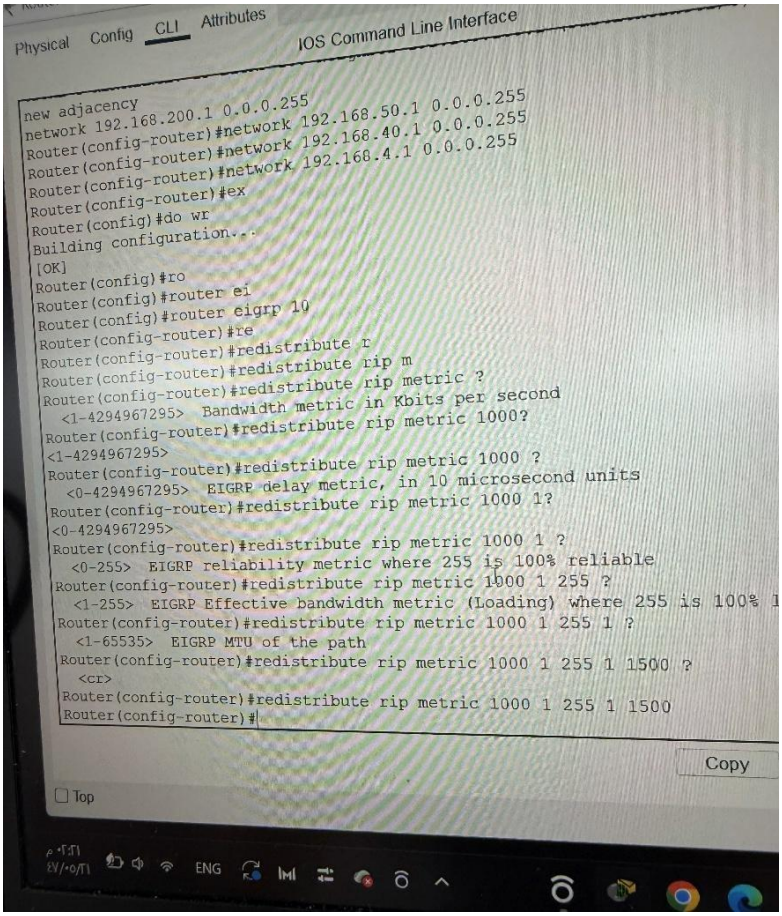
Router RIP Routing Configuration

The **Router1** showing the **RIP Routing configuration**.

The key information visible is the list of **Network Addresses** that Router1 is configured to advertise using the RIP protocol:

- 192.168.4.0
- 192.168.40.0
- 192.168.50.0
- 192.168.100.0
- 192.168.200.0

The lower CLI section confirms that networks 192.168.50.0 and 192.168.40.0 are configured under the router rip process.



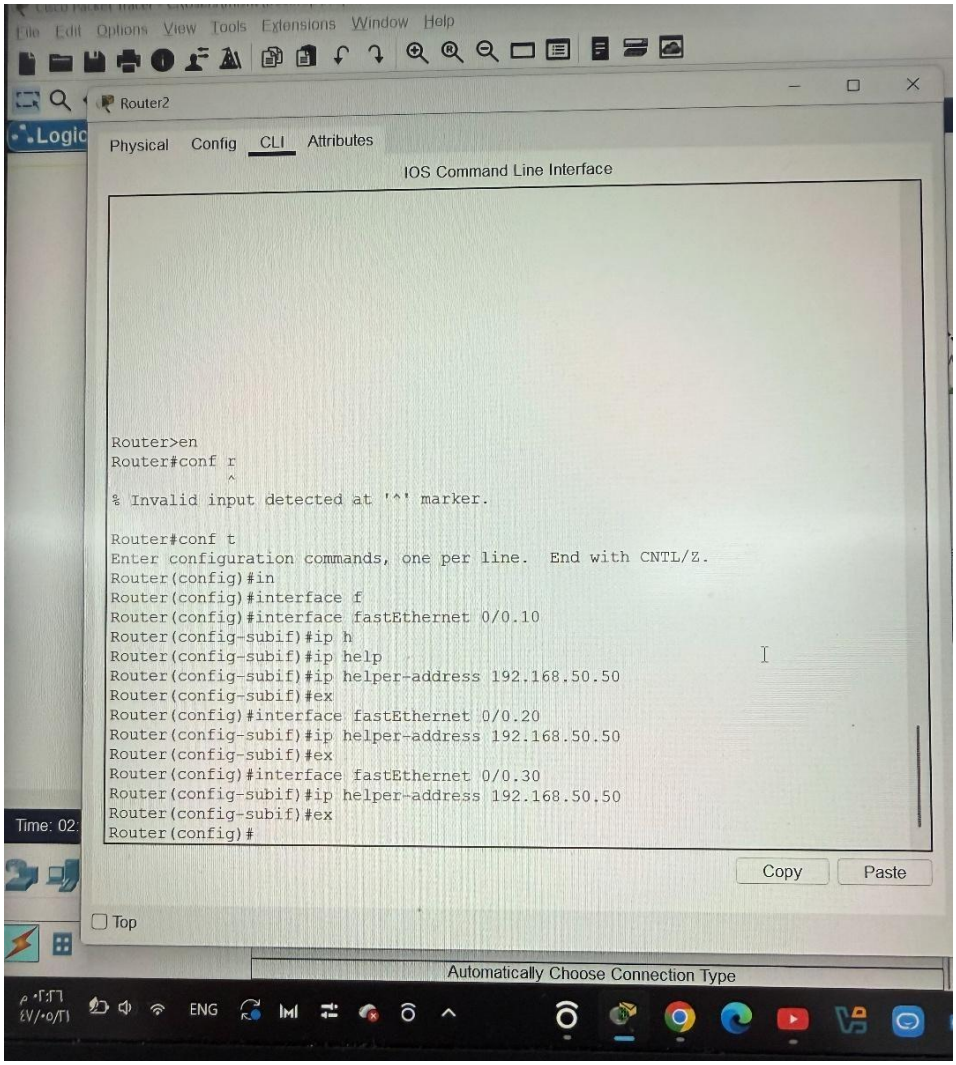
EIGRP Routing and Redistribution Configuration

The Router shows two key routing configurations:

1. **EIGRP Network Configuration:** The top section shows networks (192.168.200.1, 192.168.50.1, 192.168.40.1) being added under the router's configuration, likely for initial EIGRP setup.

2. **RIP Redistribution into EIGRP:** The main part of the screen shows the commands used to configure the **EIGRP routing protocol (Autonomous System 10)**, specifically setting up **redistribution of the RIP protocol** into EIGRP. The full EIGRP metric parameters are being defined:
   - **Bandwidth:** 1000 (Kbits per second)
   - **Delay:** 1 (10-microsecond units)
   - **Reliability:** 255 (100% reliable)
   - **Loading:** 1 (low load)
   - **MTU:** 1500

This configuration ensures that routes learned via RIP are successfully advertised to other routers running EIGRP.

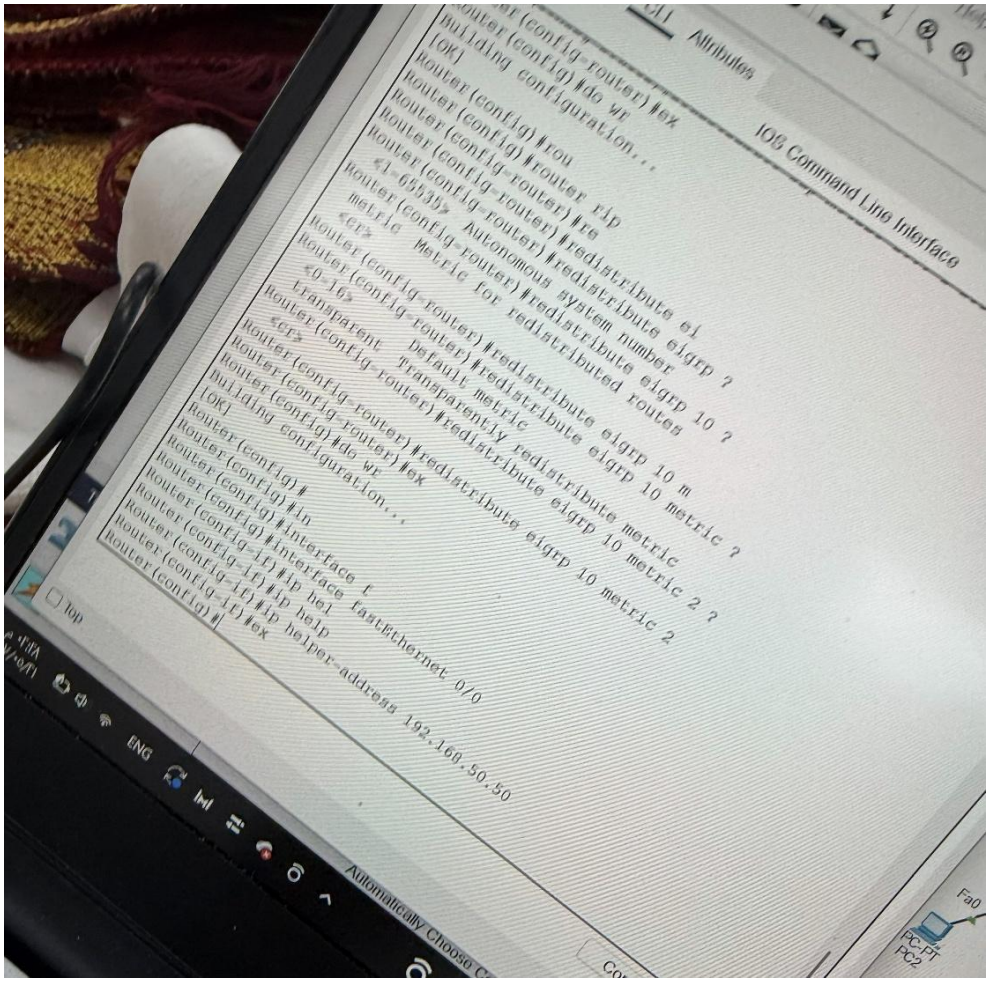


Router DHCP Helper Address Configuration

The **Router2**. It shows the configuration of the **ip helper-address** command on three different subinterfaces:

- **FastEthernet 0/0.10**
- **FastEthernet 0/0.20**
- **FastEthernet 0/0.30**

The command ip helper-address 192.168.50.50 is used to **forward DHCP broadcast requests** (and other broadcast services) received on these subinterfaces to the specified **DHCP server** located at 192.168.50.50. This is a crucial step for clients in different VLANs to obtain IP addresses when the DHCP server is in a separate network segment.

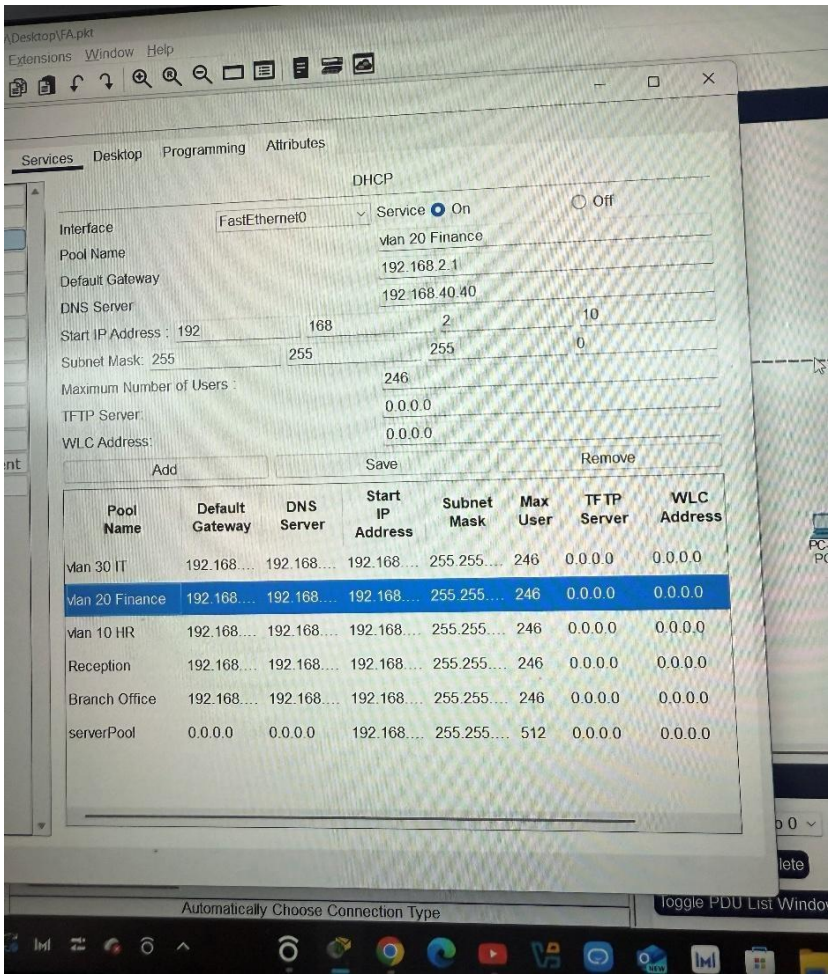


Routing Redistribution and DHCP Helper Configuration

This screenshot displays the Cisco Packet Tracer Command Line Interface (CLI) for a router, showing two distinct configuration activities:

1. **Routing Redistribution (Top Section):** Commands related to **redistributing routes** between two routing protocols, likely **RIP and EIGRP (AS 10)**. The commands attempt to set a metric for routes being redistributed from EIGRP back into RIP.

2. **DHCP Helper Address (Bottom Section):** Configuration on the physical interface **fastethernet 0/0** where the command **ip helper-address 192.168.50.50** is being applied. This forwards broadcast requests (like DHCP) from devices connected to this interface to the server at that IP address.



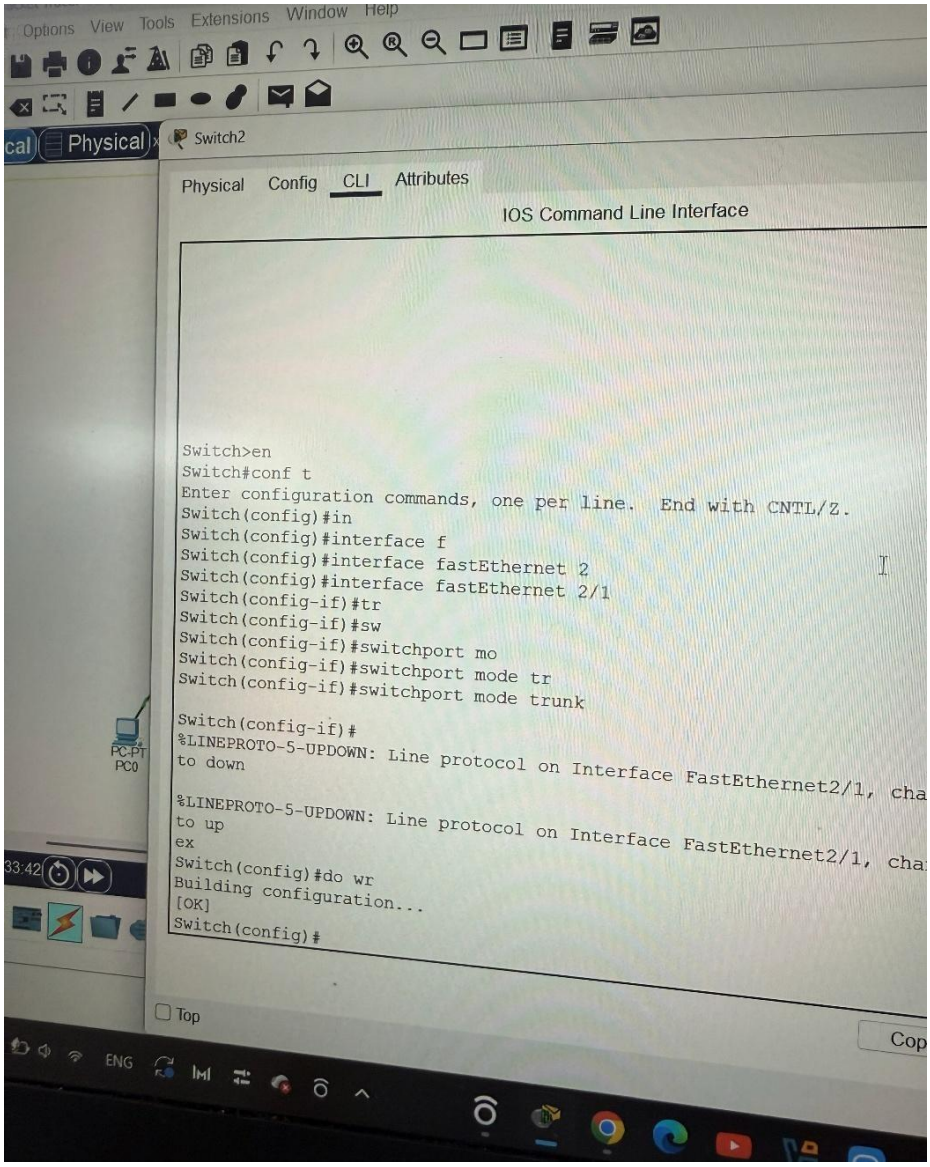
DHCP Server Configuration for Multiple Pools

The **DHCP Service configuration** interface of a device (likely a server or a router acting as a server)
The key information is the definition of multiple DHCP pools for different VLANs and departments:

- **DHCP Pools:** Pools are configured for vlan 30 IT, **vlan 20 Finance** (currently selected and detailed at the top), vlan 10 HR, Reception, Branch Office, and a serverPool.

- **vlan 20 Finance Details:**
  - **Default Gateway:** 192.168.2.1
  - **DNS Server:** 192.168.40.40
  - **Start IP Address:** 192.168.2.10
  - **Subnet Mask:** 255.255.255.0
  - **Max Users:** 246

This configuration ensures that clients in these respective VLANs can automatically receive the correct IP addressing information from the server.

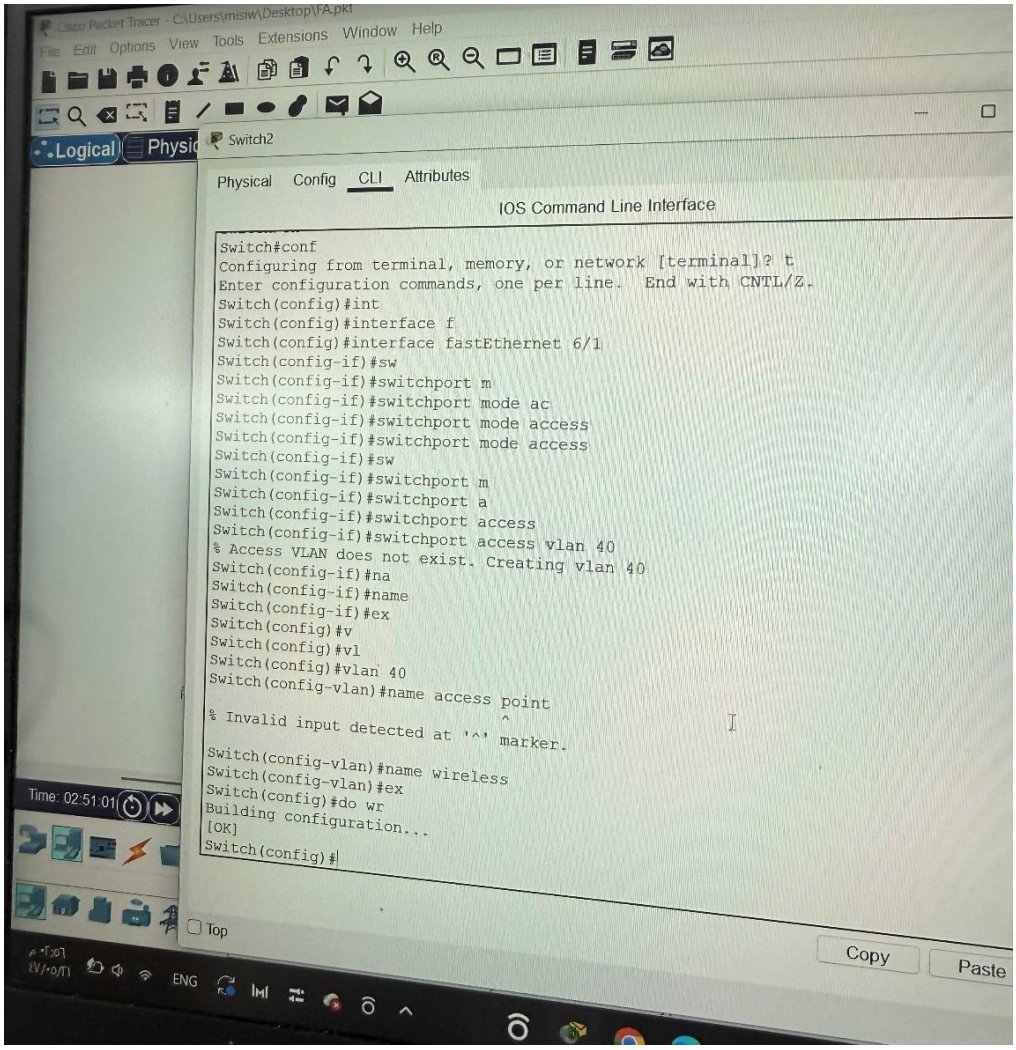


Switch Trunk Port Configuration

The **Switch2**, focusing on interface configuration.

The key action shown is the configuration of the **fastEthernet 2/1** interface as a **trunk port** using the command:

Switch(config-if)#switchport mode trunk
A trunk port is essential for carrying traffic for **multiple VLANs** over a single link, typically between a switch and another switch or a router.

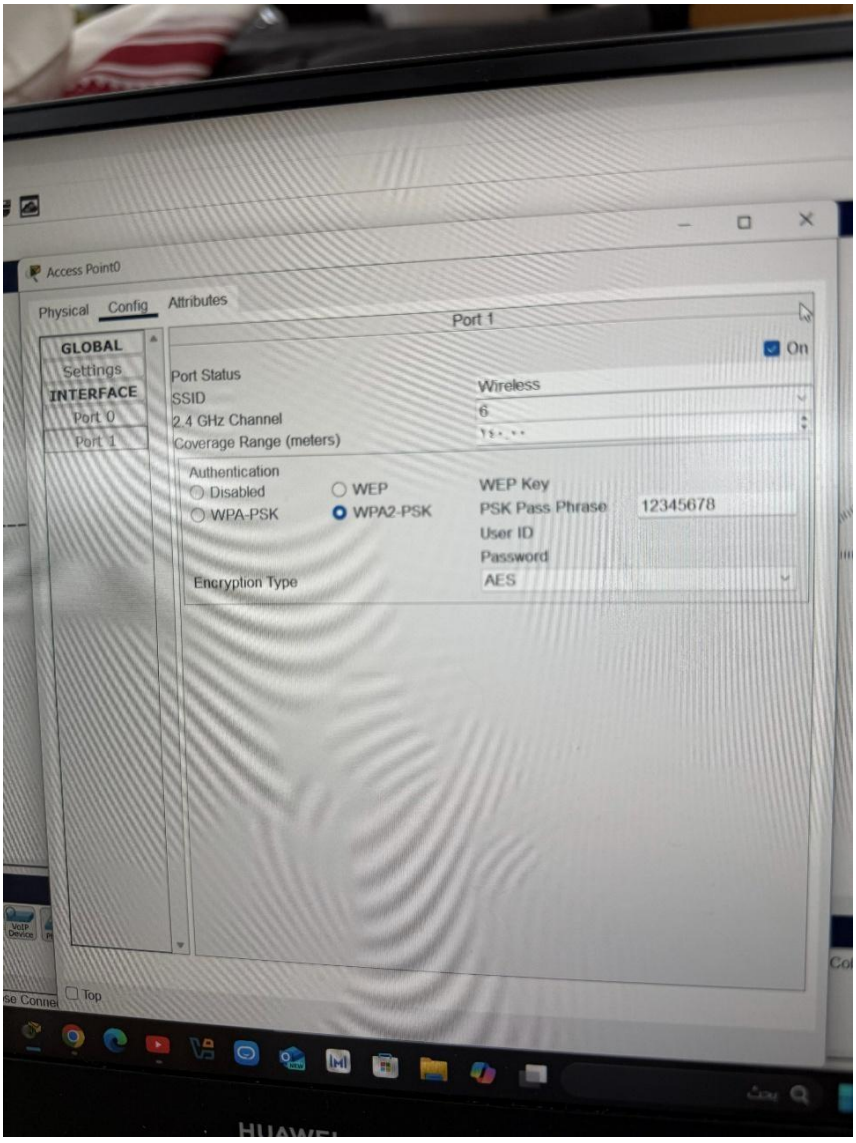


Switch VLAN and Access Port Configuration (VLAN 40)

The **Switch2**, showing configuration steps for a new VLAN:

1. **Access Port Assignment:** The physical interface **fastEthernet 6/1** is configured as an **access port** (switchport mode access).
2. **VLAN Creation and Assignment:** The interface is assigned to **VLAN 40** (switchport access vlan 40). Since VLAN 40 did not exist, the switch automatically created it.
3. **VLAN Naming Attempt:** The commands at the bottom show an attempt to name VLAN 40. The command name access point is entered, which causes an error (the ^ marker) as the correct command is typically just name [name].

This configuration isolates the device connected to Fa6/1 into VLAN 40.

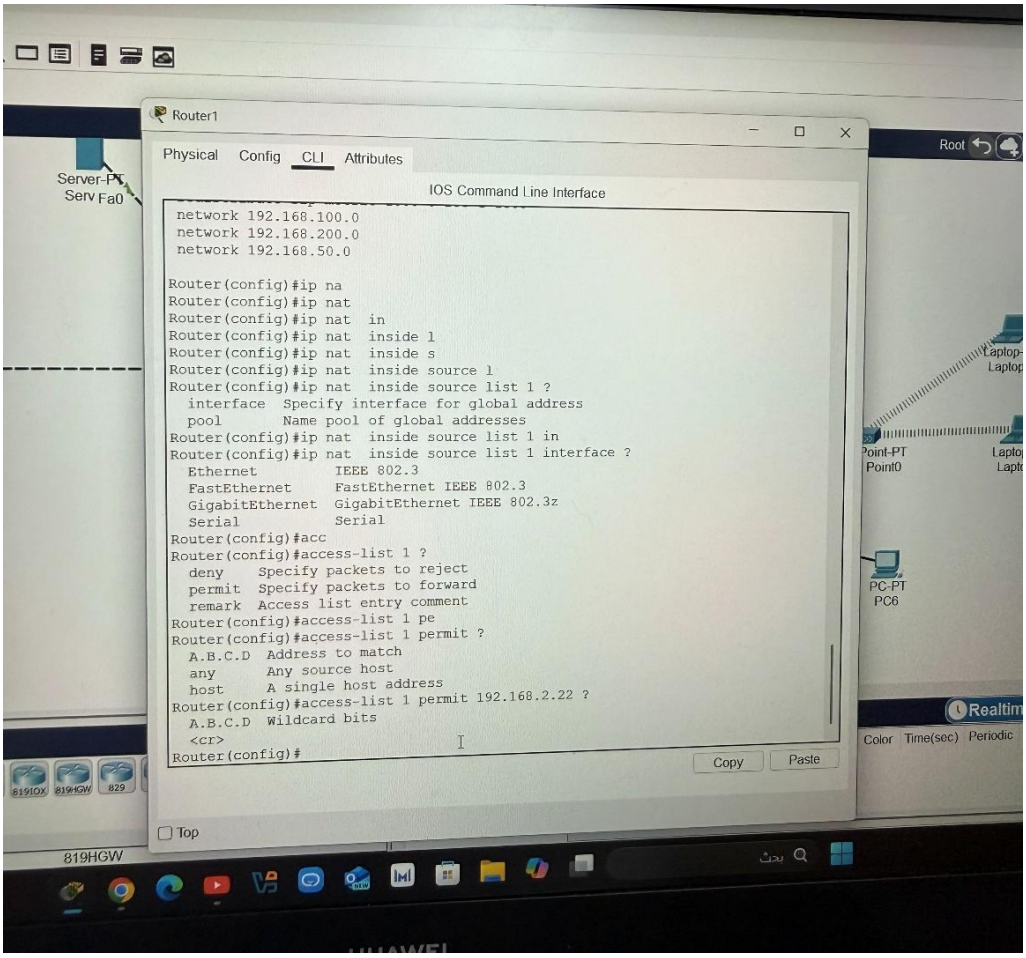


Access Point Wireless Configuration

The **Access Point0**, specifically focusing on the **Wireless (Port 1)** settings.
Key configurations shown include:

- **SSID (Service Set Identifier):** Set to **"Wireless"**.
- **Channel:** Set to channel **6** on the **2.4 GHz band**.
- **Authentication:** Set to **WPA2-PSK** (Pre-Shared Key).
- **PSK Pass Phrase:** The password is set to **12345678**.
- **Encryption Type:** Set to **AES**.

This configuration establishes the security and basic parameters for the wireless network broadcast by the Access Point.



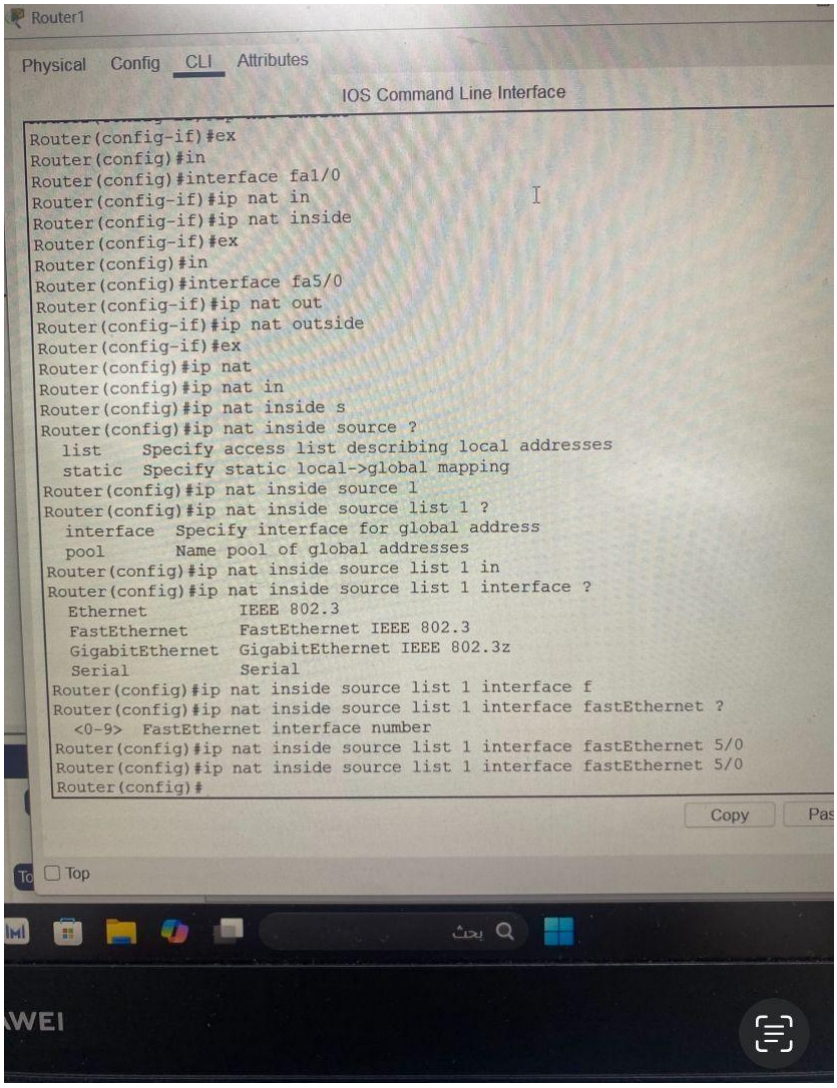
Router NAT and Access List Configuration

The **Router1**, showing commands related to Network Address Translation (NAT) and Access Control Lists (ACLs).

The key configurations visible are:

1. **NAT (Network Address Translation):** The router is being configured for **NAT Overload (PAT)** using an **inside source list 1** and an **interface** for the global address (likely the exit interface). Commands like ip nat inside and ip nat outside are being prepared.
2. **Access List (ACL):** The commands at the bottom show the start of creating an **Extended Access List 1** (access-list 1 permit...) to define traffic that should be permitted or denied, specifically mentioning an IP address 192.168.2.22.

This is a configuration typically used to secure a network and allow multiple private IP addresses to share a single public IP address.

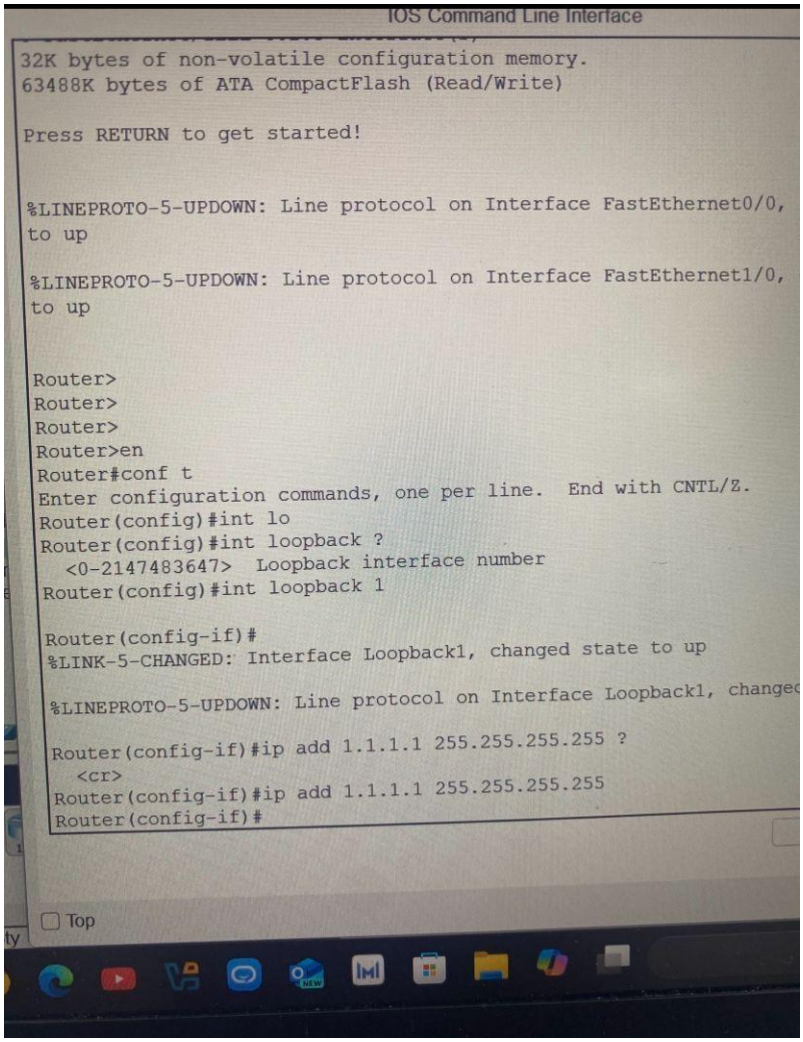


Router NAT Configuration (Router1)

The Router1, showing detailed configuration steps for Network Address Translation (**NAT**).
The key configurations visible are:

- Interface Roles: Defining the NAT inside interface (**FastEthernet 1/0**) and the NAT outside interface (**FastEthernet 5/0**) using the commands ip nat inside and ip nat outside.
- NAT Overload (**PAT**) Configuration: The interactive commands show the router being configured for **NAT** Overload (Port Address Translation), specifically using Access List 1 (ip nat inside source list 1) and the outside interface FastEthernet 5/0 for the global address pool. This allows multiple internal devices to share the public IP address of the Fa5/0 interface.

This is a crucial step for enabling internal devices with private IP addresses to communicate with the internet.



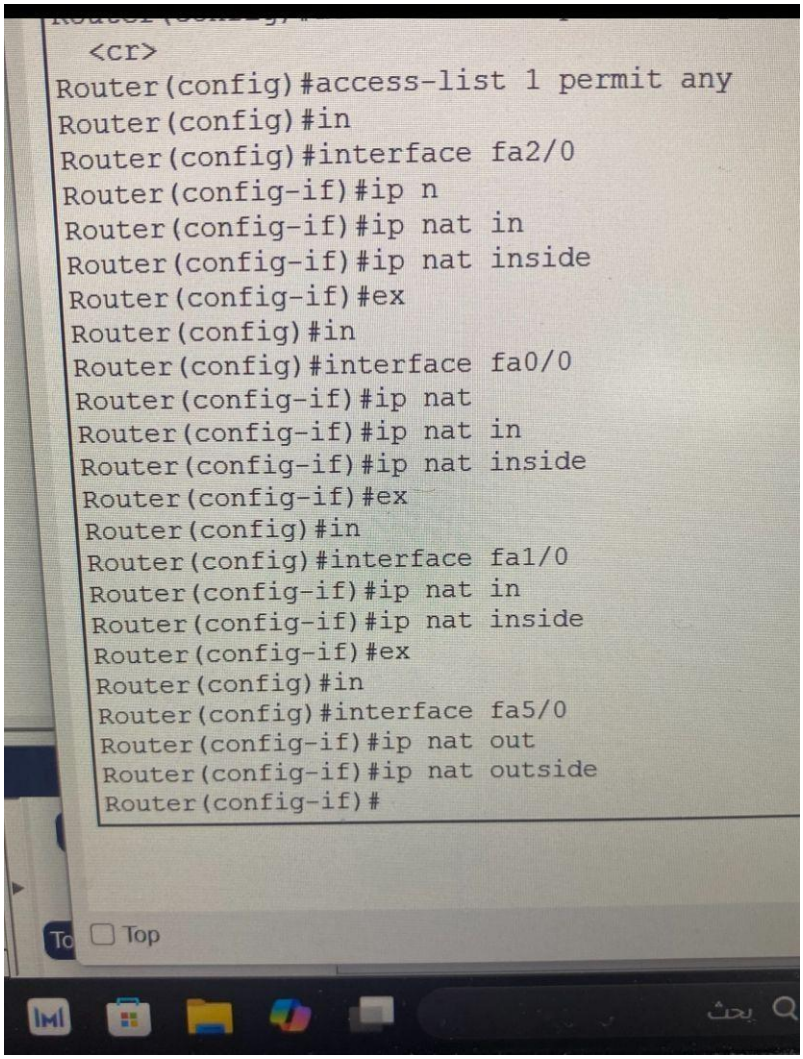
Router Loopback Interface Configuration

The router, showing the configuration of a **Loopback interface**.

The key information is:

- **Physical Interface Status:** The top messages indicate that the physical interfaces FastEthernet0/0 and FastEthernet1/0 are up.
- **Loopback Creation:** The commands show the creation of **interface loopback 1**.
- **IP Address Assignment:** The Loopback interface is assigned the IP address **1.1.1.1** with a subnet mask of **255.255.255.255**.

Loopback interfaces are commonly used for tasks like management, routing protocol configuration, or acting as a stable Router ID.



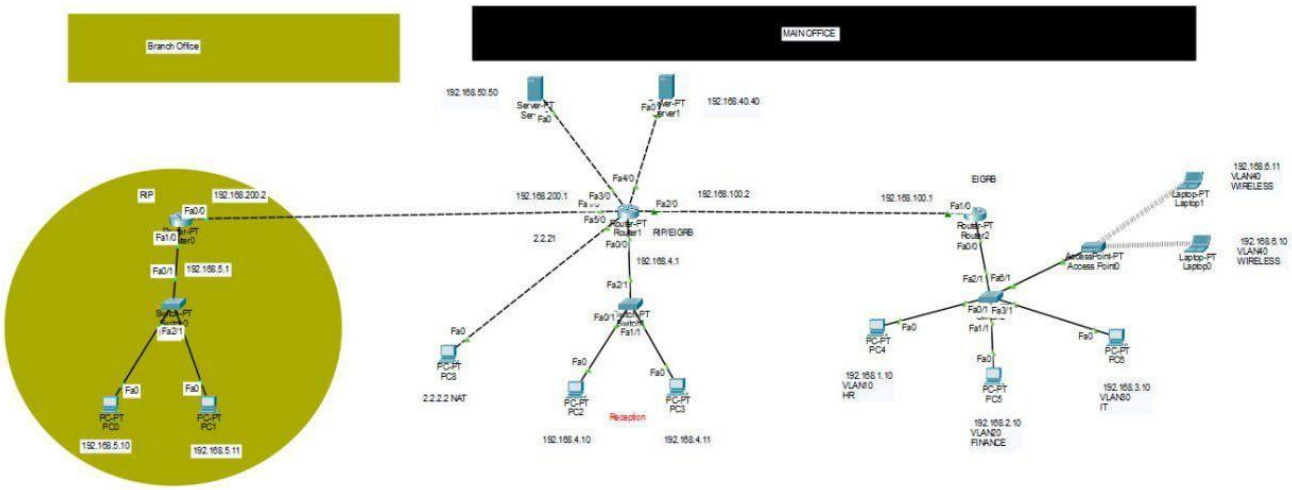
Router Multiple Interface NAT Configuration

The router, showing the configuration of Network Address Translation (NAT) roles on multiple interfaces.

The key configuration visible is the designation of:

- **NAT Inside Interfaces:** FastEthernet 2/0, FastEthernet 0/0, and FastEthernet 1/0 are all configured as NAT **inside** interfaces using the command ip nat inside. These interfaces face the local (private) network.
- **NAT Outside Interface:** FastEthernet 5/0 is configured as the NAT **outside** interface using the command ip nat outside. This interface faces the external (public) network, like the internet.
- **Access List:** An **Access List 1** is also being configured with the command access-list 1 permit any, which is likely used to define the internal traffic permitted to be translated.

This setup is used when a router is connecting multiple separate internal networks to a single external network.



The final Project Topology Diagram

The designed to demonstrate several networking concepts.

The diagram is logically segmented into three main areas:

1. **Branch Office (RIP Domain - Left):**
   - Uses the **RIP** routing protocol.
   - Includes a **Router (Router1)** connecting to a local network (192.168.5.0/24) and other PCs.
   - The link to the main office uses the 192.168.200.0/24 network.
2. **Main Office (EIGRP Domain - Right):**
   - Uses the **EIGRP** routing protocol.
   - Includes a **Router (Router2)** configured for Inter-VLAN Routing (Router-on-a-Stick, connecting to a Switch via Fa1/1).
   - The local network is segmented into **VLANs (10 HR, 20 Finance, 30 IT, and Wireless)**.
3. **Core/WAN (Center):**
   - Connects the two main domains via **Router0 (the distribution router)**.
   - The core includes a server network (192.168.90.50), a DNS server network (192.168.40.40), and a dedicated NAT subnet (2.2.2.0/24).
   - This section is likely where **Redistribution (RIP to EIGRP)** is configured on one of the central routers.

This topology is a classic example used for configuring and testing routing protocols, Inter-VLAN routing, NAT, and DHCP services.